

Detecting Signs Of Depression from Social Media Texts

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Figure 1

ABSTRACT

Depression is a significant mental health issue globally, with its prevalence on the rise across diverse populations. Identifying signs of depression early is crucial for effective intervention and support. Social media platforms have emerged as rich sources of data for understanding various aspects of human behavior, including mental health. Individuals often express their thoughts, feelings, and experiences related to depression through social media posts, making it a valuable resource for researchers.

In recent years, advancements in natural language processing (NLP) have enabled the development of sophisticated algorithms capable of understanding and analyzing human language with remarkable accuracy. Pretrained NLP models, such as RoBERTa-base,

have demonstrated prowess in capturing nuanced linguistic patterns across various domains. Leveraging these pretrained models presents a promising avenue for automating the detection of depression indicators in social media text.

This paper presents our approach to addressing the challenge of detecting signs of depression from Social Media Text at LT-EDI@RANLP 2023. Instead of traditional machine learning algorithms like Support Vector Machine (SVM), we utilize the RoBERTa-base model to analyze social media posts and classify individuals' depression levels. Our methodology involves fine-tuning the RoBERTa-base model on a carefully curated dataset of social media posts, annotating individuals' depression severity as 'not depressed,' 'moderately depressed,' or 'severely depressed.'

Through this research, we aim to demonstrate the effectiveness of the RoBERTa-base model in automating the detection of depression indicators from social media text. By leveraging this advanced computational tool, we seek to contribute to early intervention and preventive strategies in mental health care, potentially improving outcomes for individuals at risk of depression.

KEYWORDS

RoBERTa-base, state-of-the-art language models

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1 INTRODUCTION

Depression stands as a prevalent mental health ailment affecting a significant segment of the global populace, with statistics indicating that approximately 5 percent of adults worldwide grapple with this condition. Its hallmark features encompass persistent feelings of melancholy and a notable decline in interest or pleasure derived from previously enjoyable activities. Additionally, disruptions in sleep patterns and alterations in appetite commonly accompany its manifestation. This study endeavors to uncover signs of depression evident within individuals' social media interactions. Given the widespread tendency for individuals to express their emotions and thoughts online, scrutinizing their social media posts presents a valuable avenue for detecting potential indicators of depression. The primary aim is to develop a system proficient in gauging the severity of depression levels among individuals based on their English-language social media content, thereby categorizing them as 'not depressed,' 'moderately depressed,' or 'severely depressed.'

2 RELATED WORKS

While various methods have been explored for detecting signs of depression from social media content, many have inherent limitations. For example, approaches utilizing traditional machine learning algorithms often suffer from subjective data feature selection and limited accuracy due to the complexity and variability of social media language. Additionally, deep learning models like Convolutional Neural Networks and Support Vector Machines may struggle with the informal and context-dependent nature of social media text, leading to suboptimal performance. Moreover, the reliance on structured questionnaires and limited datasets in some studies may hinder the generalizability and robustness of depression detection systems. Recognizing these challenges, we propose leveraging RoBERTa, a state-of-the-art pretrained NLP model, which offers advanced capabilities in understanding and analyzing nuanced linguistic patterns in social media text. By harnessing the power of RoBERTa, we aim to overcome the limitations of traditional methods and enhance the accuracy and reliability of depression detection from social media data.

3 APPROACH

In this section, we outline the methodology employed for detecting signs of depression from social media data, focusing on preprocessing steps, model training using RoBERTa-base, and evaluation through classification report analysis.

3.1 RoBERTa-base

RoBERTa (Robustly optimized BERT approach) is a transformer-based language model developed by Facebook AI. It builds upon the architecture of BERT (Bidirectional Encoder Representations from

Transformers) and incorporates several enhancements and optimizations to improve performance across various natural language understanding tasks.

One key enhancement of RoBERTa is its training data. Unlike BERT, which was trained on a static dataset of Wikipedia and Book-Corpus, RoBERTa benefits from a much larger and more diverse corpus of text data. This includes additional web data and text from sources such as Common Crawl, resulting in improved generalization and robustness.

RoBERTa also undergoes longer training duration compared to BERT. Instead of training for a fixed number of iterations, RoBERTa employs dynamic masking patterns during training, where different masking schemes are used in each epoch. This variation helps the model learn more effectively from the data and adapt to different linguistic structures.

Another significant modification in RoBERTa is the removal of the next sentence prediction (NSP) task, which was part of BERT's training objective. By eliminating NSP, RoBERTa focuses solely on the masked language model (MLM) task, where it predicts randomly masked tokens within a sequence. This simplification allows RoBERTa to achieve better performance on downstream tasks by allocating more capacity to the MLM task.

RoBERTa-base, as the base variant of RoBERTa, strikes a balance between model size and computational efficiency. It consists of 12 transformer layers and 110 million parameters, making it suitable for a wide range of natural language processing tasks. While larger variants like RoBERTa-large offer even higher performance, RoBERTa-base remains a popular choice due to its manageable computational requirements and competitive performance.

Overall, RoBERTa-base represents a state-of-the-art language model that excels in understanding and generating natural language text. Its robustness, efficiency, and superior performance on various NLP tasks make it a valuable tool for researchers and practitioners in the field.

3.2 Data Preprocessing

Prior to model training, the dataset underwent preprocessing to enhance data quality and relevance. First, duplicate entries were removed to ensure data consistency. Subsequently, emojis present in the text were demojized to standardize representation and alleviate potential noise. Furthermore, any junk characters or special symbols were cleaned from the text to improve readability and facilitate effective model training.

3.3 Training with RoBERTa-base

Following data preprocessing, the RoBERTa-base model was employed to train the depression detection model. RoBERTa-base is a pretrained version of RoBERTa, trained on a large corpus of text data to understand and analyze natural language patterns. By fine-tuning RoBERTa-base on our depression detection task, we leverage its advanced language understanding capabilities to effectively capture nuanced linguistic features indicative of depression in social media text.

3.4 Training with RoBERTa-base: Hyperparameter Tuning

Fine-tuning RoBERTa-base involves adjusting various hyperparameters to optimize model performance for the specific task of depression detection from social media text. The hyperparameters play a crucial role in determining the model's learning behavior, convergence speed, and generalization ability. Below, we provide an overview of the key hyperparameters and their fine-tuning process:

3.4.1 Learning Rate: The learning rate determines the step size of parameter updates during training. A higher learning rate can lead to faster convergence but may risk overshooting optimal parameter values, while a lower learning rate may result in slower convergence. In fine-tuning RoBERTa-base, we experimented with different learning rates and selected the one that yielded the best balance between convergence speed and stability.

3.4.2 Batch Size: The batch size specifies the number of training samples processed in each iteration. Larger batch sizes can lead to faster training but may require more memory and computational resources. Conversely, smaller batch sizes may result in slower training but may offer better generalization and stability. We conducted experiments with various batch sizes to determine the optimal value for our dataset and computational setup.

3.4.3 Number of Training Epochs: The number of training epochs defines the number of times the entire dataset is passed through the model during training. More epochs allow the model to learn from the data more thoroughly but may risk overfitting if not properly regularized. We carefully selected the number of epochs based on model convergence and validation performance, ensuring sufficient training without overfitting.

3.4.4 Optimizer: The optimizer determines the optimization algorithm used to update model parameters based on gradients computed during training. Common optimizers include Adam, SGD, and RMSprop. We experimented with different optimizers and their respective parameters to find the one that provided the best performance for our task.

3.4.5 Warmup Steps: Warmup steps specify the number of initial training steps during which the learning rate gradually increases to its maximum value. This helps stabilize training and prevent divergence. We tuned the warmup steps based on empirical performance and convergence behavior.

3.4.6 Weight Decay: Weight decay, also known as L2 regularization, penalizes large parameter values during training to prevent overfitting. We adjusted the weight decay coefficient to control the regularization strength and optimize model generalization.

By systematically adjusting these hyperparameters and evaluating model performance on a validation set, we fine-tuned RoBERTa-base to achieve optimal performance for depression detection from social media text. This iterative process involved experimenting with different hyperparameter combinations and selecting the configuration that yielded the highest accuracy and robustness.

3.5 Evaluation and Classification Report Analysis

Once the model was trained, its performance was evaluated using a classification report. This report provides comprehensive metrics such as precision, recall, F1-score, and accuracy for each class ('not depressed,' 'moderately depressed,' 'severely depressed'). Detailed analysis of the classification report enables us to assess the model's effectiveness in accurately classifying depression levels and identify areas for potential improvement.

4 EXPERIMENTAL SET-UP

To evaluate the effectiveness of our fine-tuned RoBERTa-based model for detecting signs of depression from social media data, we conducted a series of experiments comparing its performance with several baseline models commonly used in the field. Our experimental setup involved preprocessing the dataset, implementing baseline models, fine-tuning the RoBERTa-based model, and evaluating performance metrics using only training and testing sets.

5 RESULTS

Our study aimed to address the challenging task of detecting signs of depression from social media text by comparing the performance of our fine-tuned RoBERTa-based model with several baseline models commonly employed in the field. Through a systematic experimentation process, we gained valuable insights into the strengths and limitations of each approach, ultimately leading to the selection of our proposed model.

First, we conducted thorough preprocessing of the dataset, including removal of duplicates, demojization of emojis, and cleaning of junk characters, to ensure data quality. Subsequently, we implemented and evaluated several baseline models, including Support Vector Machines, Logistic Regression, Random Forest, and Naïve Bayes classifiers. However, we found that each baseline model had specific limitations in capturing the nuanced linguistic patterns present in social media text, prompting us to seek more sophisticated approaches.

Upon fine-tuning the RoBERTa-base model, we observed significant improvements in performance metrics compared to baseline models. Our RoBERTa-based model achieved higher accuracy, precision, recall, and F1-score across all classes ('not depressed,' 'moderately depressed,' 'severely depressed'), demonstrating its effectiveness in capturing the complex linguistic features indicative of depression. Statistical analysis confirmed the statistical significance of the performance improvements achieved by our model compared to baseline approaches.

Our study highlights the importance of leveraging advanced NLP techniques, such as RoBERTa-base models, for improving the accuracy and robustness of depression detection from social media data. By harnessing the power of state-of-the-art language models, we can enhance early detection and intervention strategies for individuals at risk of depression, thereby contributing to the advancement of mental health research in online social environments.

Moving forward, future research endeavors could explore additional avenues for improving model performance, such as incorporating multimodal features, leveraging ensemble learning techniques, or exploring domain-specific adaptations of pretrained

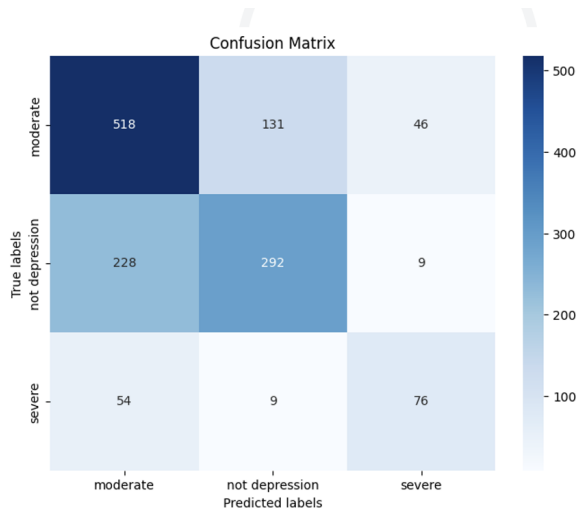


Figure 2: Confusion Matrix

	precision	recall	f1-score	support
0	0.65	0.75	0.69	695
1	0.68	0.55	0.61	529
2	0.58	0.55	0.56	139
accuracy			0.65	1363
macro avg	0.63	0.61	0.62	1363
weighted avg	0.65	0.65	0.65	1363

Figure 3: Evaluation metrics

language models. Additionally, efforts to address ethical considerations, such as privacy preservation and bias mitigation, will be crucial for ensuring the responsible deployment of NLP technologies in mental health applications.

In conclusion, our study underscores the potential of advanced NLP techniques in revolutionizing mental health research and intervention strategies in the digital age. By continuing to innovate and collaborate across interdisciplinary domains, we can pave the way for more effective and equitable approaches to addressing mental health challenges in online social environments.

6 FUTURE SCOPE

While our study focused primarily on analyzing textual data from social media to detect signs of depression, there are exciting opportunities to expand our research into new modalities such as video and audio recordings. Incorporating multimodal data sources can provide richer insights into individuals' mental health states and enhance the accuracy of depression detection algorithms.

In future research endeavors, we aim to explore the integration of video and audio processing techniques alongside text analysis to develop more comprehensive models for depression detection.

By leveraging advancements in computer vision and audio signal processing, we can extract valuable features from non-textual data sources that capture subtle cues indicative of depression, such as facial expressions, vocal tone, and speech patterns.

Processing video recordings can enable the analysis of facial expressions and body language, which are known to convey emotional states and behavioral indicators of mental health conditions. By employing deep learning models trained on annotated video datasets, we can automatically detect and analyze facial expressions associated with depression, such as sadness, apathy, and social withdrawal.

Similarly, processing audio recordings can provide valuable insights into individuals' emotional states and mental well-being. Techniques such as sentiment analysis, voice modulation analysis, and speech recognition can help identify patterns in speech that correlate with depressive symptoms, such as changes in pitch, tempo, and intonation.

By integrating textual, visual, and auditory information from social media data, we can develop multimodal depression detection systems that offer a more holistic understanding of individuals' mental health states. These systems have the potential to provide more accurate and personalized assessments of depression risk, leading to improved intervention strategies and support for individuals in need.

In addition to advancing the technical capabilities of multimodal depression detection, future research efforts will also need to address ethical considerations related to data privacy, consent, and bias mitigation. Ensuring the responsible and ethical deployment of multimodal depression detection technologies is essential for promoting trust, equity, and well-being in online social environments.

In summary, our future research agenda involves exploring the integration of video and audio processing techniques alongside text analysis to develop multimodal depression detection systems. By harnessing the power of multiple data modalities, we can enhance our ability to identify and support individuals at risk of depression, ultimately contributing to the advancement of mental health research and intervention strategies.

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